

Magnet Therapy

A Billion-dollar Boondoggle

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About a billion dollars a year is now spent on “magnet therapy,” which is claimed to eliminate many symptoms and diseases. Basic scientific principles indicate that all of this money is wasted.

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About a year ago Leonard Finegold at Drexel University and I decided to look into the controversial field (no pun intended) of “magnet therapy.” As a physics professor, Finegold knows a bit about magnets and magnetic fields. As a physician and former research chairman, I know a bit about therapy and medical research. Perhaps a physicist and a physician could shed some light on this interesting topic. We knew that magnets were touted as a treatment for many medical conditions and we knew that they were popular. But we were both quite surprised to learn just how popular they are. In the U.S.A., their annual sales are estimated at \$300 million (Brody 2000), and globally more than a billion dollars (Weintraub 1999). You can get a rough idea of the magnitude of the magnet healing industry by doing a Google search for *magnet healing*. A search in January 2006 yielded 459,000 Web pages, many of them claiming that magnets have almost miraculous healing power. Do they? Professor Finegold and I reviewed the literature on magnet therapy and found very little supporting evidence. An abbreviated version of our review was recently published in the *British Medical Journal* (Finegold and Flamm 2006). What follows are a few comments on the magnet healing industry, a brief synopsis of our *BMJ* paper, and a look at magnet therapy from a theoretical point of view.

Magnet Therapy Is Big Business

If you try the Internet searching experiment described above you will notice that in addition to almost half a million pages dealing with magnet therapy, Google automatically provides a list of “sponsored links.” Your computer screen will fill with the names of companies that have paid to help you find their site. What do these sites offer? If you click on www.magnetsandhealth.com, you will learn that “magnets help to flush out toxins in our body” and that “our magnet products have both beauty and health benefits, they increase blood flow and they increase the oxygen level in the body.” Really? They also point out that their magnets are small and mobile, which “allows you to heal the ailments of yourself and your family without having major interruptions in your life and routine. You also get all the benefits without having to go for expensive sessions with a magnetic therapist or having to take expensive courses of drugs which can also have harmful side effects.” The message seems quite clear: Why bother with doctors and medicines when magnets are safe and effective?

Another of the scores of sponsored links is www.magnetic-therapymagnets.com. This site is interesting because, in addition to selling dozens of magnetic healing devices for humans, it doesn't forget about Fido. For only \$11.95 plus shipping they will send you an amazing pet collar that will “keep your cat or dog in excellent health and vitality with constant magnetic therapy.” My wife and I are now kicking ourselves for spending thousands of dollars on veterinary care over the past several years. If we had only bought that collar!

I don't mean to pick on these two companies or imply that their claims are any more outrageous than any others. In fact, there are now hundreds of companies selling similar devices and making similar claims.

Among the companies touting magnet therapy I was surprised to find the Sharper Image, a seemingly reputable outfit. They offer a device called a “Dual-Head Personal Massager with Magnetic Therapy.” It is somewhat phallus-shaped, small enough to fit in a purse, and claims to be a “discreet personal massager with two independent vibrating heads.” That certainly seems enticing enough, but they insist that it does far more than your average vibrator. “A smaller pinpoint node enhances its massage with magnetic therapy for focused treatment.” Hmm . . . magnet therapy for focused treatment.

Some companies actually claim that their magnets prevent, reverse, and cure cancer. For example, at one site (<http://azunimags.bigstep.com/item.jhtml?UCIDs=52566311230205&PRID=1169182>) purveyors of cancer-curing magnets will sell you, for only \$2,595, the “Dr. Philpott Designed and Approved Polar Power Super Bed Grid.” According to the site, “This is the strongest, deepest penetrating, permanent static magnet, bio-magnetic therapy device available anywhere that we know of. It is used in many of Dr. Philpott's magnetic research protocols for prevention and reversal of cancer and other serious disease that requires a full systemic deep penetrating treatment of the whole body.” Similarly absurd claims can be found at www.stop-cancer.com/magnets.htm.

Do the legions of magnet therapists and magnet purveyors really believe the incredible claims that they make? Are they well-meaning but misguided individuals or con artists willing to say anything to make a buck? Both types are most likely involved.

Studies on Magnet Therapy

The overall conclusion of our *BMJ* review was stated in the first sentence, “We believe there is a worldwide epidemic of useless magnet therapy” (Finegold and Flamm 2006). As you can imagine, this statement was not well-received in the magnet healing community. We found that many studies on “magnet therapy” were published in “alternative” journals as opposed to peer-reviewed medical journals. Many studies included too few patients to reach statistically significant conclusions. Others had problems with their placebo control groups. For example, study subjects realized that they were wearing a magnetic bracelet rather than a placebo bracelet when it attracted paper clips or other small metal objects. In light of the vast amounts of money spent each year on supposedly therapeutic magnets, surprisingly few legitimate randomized controlled trials have been conducted to evaluate their efficacy. An excellent critique of magnet therapy by Quackwatch founder Stephen Barrett, M.D., can be found at www.quackwatch.org/04ConsumerEducation/QA/magnet.html.

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Is Magnet Therapy Even Theoretically Possible?

In reality, many people find anecdotal reports of healing, particularly from athletes or other trusted celebrities, to be more convincing than scientific studies.

There are certainly many people touting magnet therapy. But is there, even theoretically, any way that magnets could have any healing effect? In our *BMJ* review we restricted our comments to typical magnetic devices claimed to have therapeutic value: these use “static” magnets like those used to attach paper notes to a refrigerator door. In this context, *static* means nonmoving and has nothing to do with static electricity. Moving magnets or pulsed electromagnets can create electric fields and electromagnetic radiation that could have some effect on living tissue. In contrast, a typical nonmoving magnet produces only a magnetic

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field. Is there anything in the human body that is affected by magnetic fields? Surprisingly, the answer appears to be no. This seems counterintuitive since most people know that oxygen in our blood is carried by hemoglobin and that hemoglobin contains iron. This is why iron tablets are often recommended for the treatment of anemia. However, the iron in hemoglobin is not ferromagnetic (see www.badsience.net/?p=88 and www.hfienberg.com/clips/magnet.htm). If hemoglobin contained ferromagnetic iron it would be simple to separate red blood cells from other blood cells with a magnet. Several studies have shown that static magnetic fields do not affect blood flow (see www.hfienberg.com/clips/magnet.htm and www.quackwatch.org/01QuackeryRelatedTopics/PhonyAds/florsheim.html). Perhaps more important, if hemoglobin contained ferromagnetic iron people might explode or be flung across the room when exposed to the extraordinarily powerful magnetic field of a MRI scan. For a fascinating look at things that can go wrong when ferromagnetic materials get too close to the powerful magnetic field of an MRI machine, visit <http://mripractice.tripod.com/mripractice/id69.htm> and www.simplyphysics.com/flying_objects.html.

However, for the sake of argument, what if some effect of magnets on human tissue could be demonstrated? What is the likelihood that it would be a therapeutic or healing effect? Probably

slim to nil. By analogy, consider chemical compounds. The number of known chemical compounds is on the order of ten million. However, only a handful have ever been shown to have any therapeutic effects. Yet millions are toxic. It would be most unwise to eat or drink anything found on the shelves of a typical chemistry lab. If a magnet had an effect on human tissue, there is no reason to believe that it would necessarily be a healing effect.

Moreover, even the rare chemical compound that has healing effects usually does so only in very specific dose ranges. Almost any prescription drug can harm or kill you if you ingest enough of it. If, theoretically, a magnet had some effect on human tissue and if, astoundingly, the effect was beneficial rather than toxic, would one not expect there to be an optimal dosage? Yet, advertised healing magnets vary widely in their field strength. Many magnet purveyors claim that the more powerful the magnet, the greater the healing effect. This sounds good but makes little sense. All known effective therapies—including medications, x-rays, and lasers—become toxic or damaging at high levels. Nevertheless, the “magnet therapist” who debated me on BBC radio immediately after our paper was published chided me for not understanding that some magnet healers fail because they don’t use strong enough magnets. She was so convincing that I think she actually believes this. The BBC radio host made a point of stating that the magnet therapist was “certified.” By whom, I wondered? The Intergalactic Association of Magnetic and Crystal Healers? After the show my colleague Professor Finegold, who was raised in the United Kingdom, informed me that the word “certified” has a derogatory mental health connotation in the UK. Perhaps the BBC host was not flattering my opponent.

Some magnet advocates contend that no one has conclusively proven that magnets cannot heal. Of course, they have it backwards. When it comes to healing, the burden of proof is on the seller, not the buyer. One is supposed to prove that a therapy works before marketing it to the public. If this were not true, medical companies could save billions by selling all sorts of untested drugs and devices. In reality, the government insists that every medicine and therapeutic device be meticulously tested for both safety and efficacy. This protective system generally works and only rarely do unsafe or ineffective products slip through and reach the public. Sadly, it seems that no such protective laws exist for magnets, crystals, amulets, magic potions, or other claimed miracle cures.

Finally, in the firestorm of criticism that followed the publication of our *BMJ* article, a frequent complaint was that I don’t have an “open” mind. It might be more fair to say that my mind is open—but not to nonsense. If properly conducted research demonstrates a genuine healing effect of static magnets, I will cheerfully incorporate magnet therapy into my clinical practice. Until that time, I hope that parents will take their sick children to evidence-based physicians rather than “certified” magnet healers.

Notes

1. Brody, J. 2000. Less pain: Is it in the magnets or in the mind? *New York Times*, November 28: F9.
2. Weintraub, M. 1999. Magnetic bio-stimulation in painful diabetic peripheral neuropathy: A novel intervention—a randomized, double-placebo crossover study. *American Journal of Pain Management* 9: 8–17.
3. Finegold, L., and B.L. Flamm. 2006. Magnet therapy: Extraordinary claims, but no proved benefits. *British Medical Journal* 332: 4. □